Marmara University
Engineering Faculty, Electrical and Electronics Engineering
EE4077 Fundamentals of Machine Learning, Final Exam, Spring, 2025

1. Consider the following two-dimensional six observations

$$\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}0\\2\end{bmatrix}, \begin{bmatrix}2\\0\end{bmatrix}, \begin{bmatrix}1\\3\end{bmatrix}, \begin{bmatrix}3\\1\end{bmatrix}, \begin{bmatrix}3\\3\end{bmatrix}$$

(a) [4 pt] Compute the mean of the sample points and write the centered observation matrix **X**.

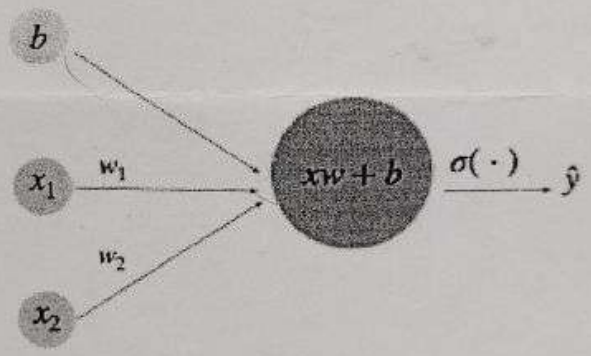
(b) [10 pt] Find all the principal components of this sample. Write them as unit vectors.

(c) [2 pt] Which of those two principal components would be preferred if you use only one?

(d) [2 pt] What information does the PCA algorithm use to decide that one principal component is better than another?

(e) [2 pt] From an optimization point of view, why do we prefer that one?

2. Consider the following simple neural network.



The network receives two binary inputs x_1, x_2; computes $x_1 \times w_1 + x_2 \times w_2 + b$ followed by a threshold activation function, namely

$$\sigma(z) = \begin{cases} 1, & z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

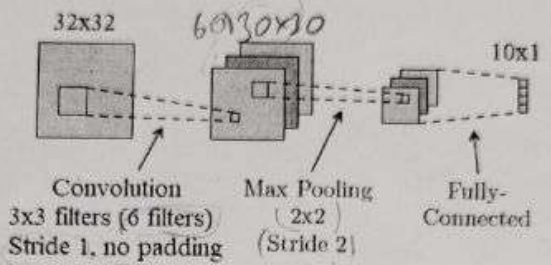
(a) [8 pt] If $b = -10$, provide values for the weights w_1, w_2 allowing us to compute the logical AND function (which is 1 if and only if two inputs are 1, otherwise 0)

(b) [8 pt] If $b = -10$, provide values for the weights w_1, w_2 allowing us to compute the logical OR function (which is 0 if and only if two inputs are 0, otherwise 1)

(c) [4 pt] Is it possible to implement the logical XOR function (which is 0 if both inputs are same, otherwise 1) with this network?

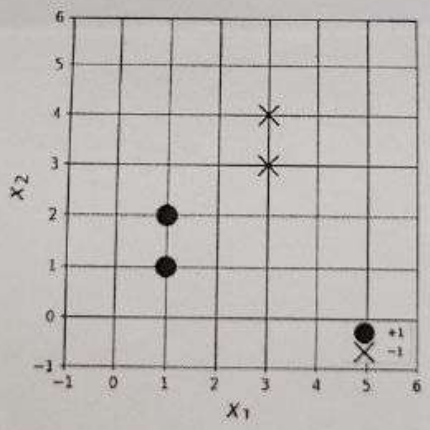
3. Below is a diagram of a small convolutional neural network that converts a 32x32 image into 10 output values. The network has the following layers/operations from input to output: convolution with 6 filters of size 3x3, max pooling, ReLu, and finally a fully connected layer (no bias/offset parameters) with 10 outputs. Answer the following questions about this network.

(a) [5 pt] How many weights in the convolutional layer do we need to learn?
(b) [5 pt] How many ReLu operations are performed on the forward pass?
(c) [5 pt] How many weights do we need to learn for the entire network?
(d) [5 pt]What is the disadvantage of a fully connected neural network compared to a convolutional neural network with the same size layer



4. Below is an illustration of a two-dimensional feature space and four training points, two of class O (label $y_i = +1$) and two of class X (label $y_i = -1$). Draw the decision boundary of the maximum margin classifier (also known as a hard-margin support vector machine).

(a) [8 pt] Draw the decision boundary of the maximum margin classifier (also known as a hard-margin support vector machine).
(b) [12 pt] Compute the maximum margin classifier's optimal parameters w_1, w_2, and α.



5. Answer the following questions. Answers without justification will get no credit.
(a) [5 pt] **True/False.** CNN can be used for image captioning and language translation problems
(b) [5 pt] **True/False.** Backpropagation algorithm is used to update RNN model parameters
(c) [5 pt] **True/False.** K-means is an unsupervised machine learning algorithm.
(d) [5 pt] **True/False.** PCA was developed for clustering.

Directions:
1. Closed book, closed notes.
2. The exam is 75 minutes.
3. Justify your answers.